

Memory-Centric Agentic Inference — cycles 29-31

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Memory-Centric Agentic Inference — cycles 29-31

Abstract

Cycles 29-31 extended the production-evidence control path for the memory-centric agentic inference package. The work did not make new performance claims for Option B or Option C. It added two validated fail-closed gates that define how future production telemetry may move from signed operator evidence toward claim readiness: an operator trust-policy gate in cycle 29 and an end-to-end evidence gatechain replay in cycle 30.

Cycle 29 produced M-TRUSTPOL-1, the operator trust-policy and production signing replacement design. It defines the policy evidence required before fixture signing can be replaced by operator key-management service (KMS), hardware security module (HSM), hardware-attestation root, or operator certificate authority evidence. A complete fixture policy can reach `trust_policy_admissible=true`, but no profile grants `attestation_source_trusted`, `production_trust_established`, `production_calibrated`, `production_ready`, or claim credit. The auditor validated 23 schema rows, 1 complete profile, 11 invalid profiles, 5 key-lifecycle rows, 10 replacement-map rows, 12 policy results, 8 failure categories, 12 boundary rows, 4 traceability links, and three nonblank figures.

Cycle 30 produced M-GATECHAIN-1, an end-to-end production evidence gatechain and promotion-state replay. It composes the prior gates into a single ordered state machine from `raw_bundle_seen` through attestation, trust policy, intake custody, adapter conformance, normalization, production ingestion, security/provenance, noise floor, threshold replay, planner eligibility, final readiness, handoff traceability, and finally `production_claim_credit_allowed`. The supplied audit report found one critical defect: replay had treated a state as passed when its `state_id` appeared, even if `state_passed=false`. The fix enforces `state_passed == true`, adds an invalid-failed-attestation fixture, and verifies failed-gate quarantine. The final validated output contains 14 state rows, 14 transition rules, 26 valid fixture rows, 202 invalid fixture rows, 19 replay summaries, 17 quarantine reasons, 19 claim-credit boundary rows, 5 traceability rows, and three nonblank figures. All 19 replay paths have `production_claim_credit_allowed=false`.

Cycle 31 has supplied researcher, worker, and auditor session IDs, but no separate cycle-31

milestone, script, test, data file, markdown artifact, figure, or ledger event was found in the available workspace record. This report treats cycle 31 as a source-record gap.

The conclusion remains conservative. The campaign now has individual gate validation and an end-to-end promotion-state replay, but Option B and Option C remain validated mechanisms and contract-ready pathways, not production recommendations. Real trusted production_target telemetry with deployment-root integration is still required before DC-001/DC-002 or Option B/C claims can be promoted.

Introduction

The campaign investigates whether long-running agentic LLM inference should be designed around memory movement, placement, reuse, compression, and lifetime management rather than arithmetic throughput alone. Earlier cycles built a research package around workload taxonomy, memory-object lifetimes, synthetic cost and queueing models, executable simulators, scheduling abstractions, runtime prototypes, security/provenance gates, production-shaped telemetry contracts, deployment blueprints, and final readiness boundaries.

Cycles 29-31 focus on a narrower but important systems question: what must be true before future operator telemetry can be trusted enough to affect production claims? The answer is not just “the files exist” or “the signatures verify.” The package now distinguishes mechanical validity, policy admissibility, production trust, production ingestion, threshold replay, final readiness, and handoff traceability as separate gates.

This distinction matters for the central memory-centric architecture question. The package contains simulated and host-local evidence that motivates Option B, a memory-object-aware architecture, and Option C, a trajectory/DAG-aware architecture. But those options should not become production recommendations unless real production telemetry proves safe reuse or trajectory value under joined bytes, power, latency, tenant, topology, security, provenance, retention, verifier, interval, and noise-floor gates. Cycles 29-30 strengthen the evidence boundary that protects that conclusion.

The hardware and systems context remains the same as prior reports: modern accelerator platforms expose GPU HBM, host DRAM, interconnect, storage, and pooled-memory tiers [1]-[7]; LLM serving systems already use paged KV-cache management, prefix caching, and semantic caching [9], [12], [13]. The unresolved production question is whether agentic workloads create enough durable, reusable, branch-heavy state to justify a memory-centric infrastructure design beyond throughput-oriented serving.

Methodology

This report consolidates accepted work rather than re-auditing it. The source inventory for cycles 29-31 is:

Cycle	Milestone	Source sessions	Main artifacts
29	M-TRUSTPOL-1	researcher 2f10dd0e-410c-437f-86b7-632c6afbc58f; worker cd141b85-6781-49f4-a0b1-79f46a87b759; auditor f57a5389-0956-4b3e-b994-6fc4e5aeb85b	operator_trust_policy.md, trust-policy schema, valid and invalid policy profiles, key lifecycle matrix, attestation replacement map, policy results, failure modes, boundary rows, traceability links, three figures
30	M-GATECHAIN-1	researcher 430d6071-c8ff-4d6d-919a-8955a2de14f5; worker 7b936d68-21d3-4a35-b928-04cab99f5417; auditor d4f6088c-1a41-40ee-8003-49883038baa3	evidence_gatechain.md, state schema, transition rules, valid and invalid replay paths, replay results, quarantine reasons, claim-credit boundary, traceability matrix, three figures
31	Record gap	researcher 4e6ccae6-d9c1-468f-9c9f-f901bafabe2b; worker 58d60678-2d3b-41c2-a833-68738527836b; auditor e53d1f06-83a9-4327-b2f1-5159718cae6b	No separate cycle-31 workspace artifact or ledger event found

No callable `search_sessions` or `list_session_catalog` tool was available in this environment, and no MCP session resources were exposed. The report therefore uses the supplied session IDs as traceability anchors and relies on workspace artifacts, `promise_ledger.jsonl`, `plan_of_record.md`, `REFERENCES.md`, `MANIFEST.md`, generated CSVs, generated figures, the prior cycle report, and the supplied audit report.

Results

Context Before Cycle 29

Cycles 26-28 had completed the adapter and intake path. M-PORT-1 verified that backend-shaped adapter outputs could canonicalize aliases, declare units, align clocks and intervals, preserve tenant/security context, and stay labeled as non-production conformance evidence.

M-INTAKE-1 defined the front-door production telemetry intake bundle: manifest schema, payload inventory, checksums, join window, provenance, measurement quality, security/privacy declarations, and boundary labels.

Those gates made future telemetry structurally admissible, but they did not solve production trust. Cycle 29 addressed the missing layer between a mechanically valid test attestation envelope and a real operator deployment root.

Cycle 29: Operator Trust Policy and Signing Replacement

Cycle 29 added M-TRUSTPOL-1, the operator trust-policy gate. A trust policy is the set of controls that makes a signing or attestation mechanism usable as production evidence. In this package, it must sit between the existing test signing envelope and any future production signing deployment.

The replacement point is explicit. The fixture mechanisms `hmac_sha256_test_fixture`, `test-key-active-a`, and `test_attestation_fixture` must be replaced by an operator KMS, HSM, hardware-attestation root, or operator certificate authority. The replacement must include non-exportable custody, rotation, revocation, collector identity binding, replay protection, tenant/security binding, and auditability.

The validated policy outputs are:

Artifact	Rows	Role
<code>data/operator_trust_policy_schema.csv</code>	23	Required policy fields across trust root, key custody, collector binding, replay, audit, and tenant/security dimensions
<code>data/operator_trust_policy_profiles.csv</code>	1	Complete fixture policy profile
<code>data/operator_trust_policy_invalid_profiles.csv</code>	11	Invalid policy profiles
<code>data/operator_key_lifecycle_matrix.csv</code>	5	Required key lifecycle phases and fail-closed reasons
<code>data/operator_attestation_replacement_map.csv</code>	10	Mapping from fixture attestation fields to production replacements
<code>data/operator_trust_policy_results.csv</code>	12	Policy admissibility and blocked-reason outputs
<code>data/operator_trust_policy_failure_modes.csv</code>	8	Failure category counts
<code>data/operator_trust_policy_boundary.csv</code>	12	Boundary rows showing policy admissibility is not production trust
<code>data/operator_trust_policy_traceability_links.csv</code>	4	Links to attestation, intake custody, deployment preflight, final readiness, and handoff

The complete fixture policy can reach `trust_policy_admissible=true`. It still keeps `attestation_source_trusted=false`, `production_trust_established=false`, `production_target_granted=false`, `production_calibrated=false`, `production_ready=false`, and `claim_credit_allowed=false`.

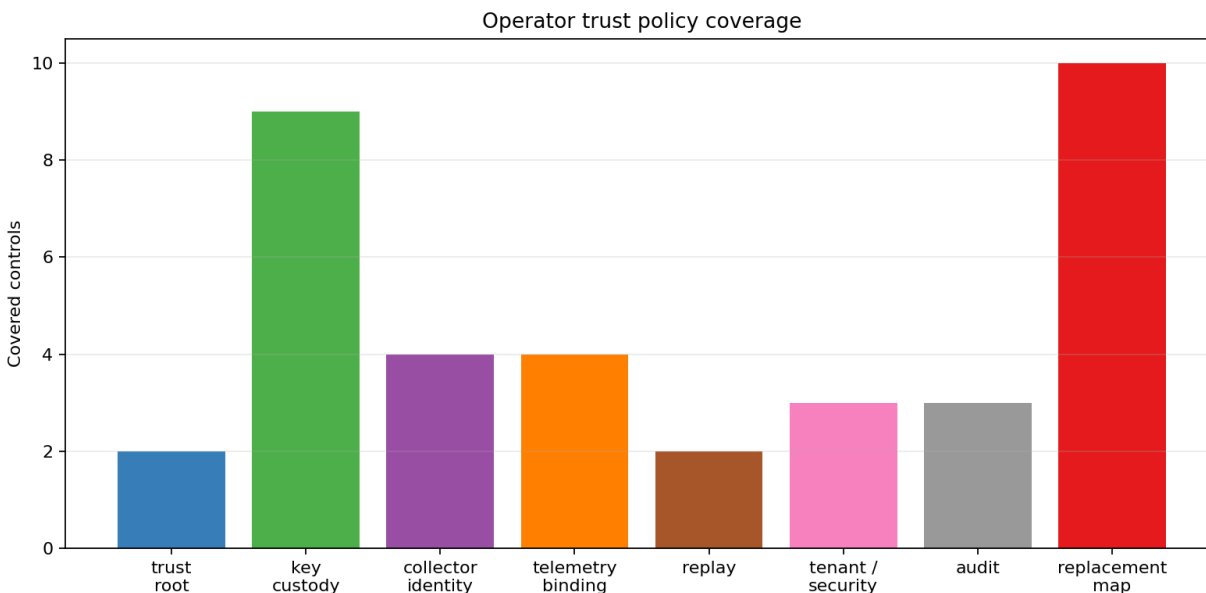


Figure 1: Coverage of required operator trust policy dimensions.

The invalid profiles fail closed for fixture HMAC presented as a production root, missing revocation, exportable production key material, unbound collector identity, missing firmware identity binding, missing replay protection, missing audit logging, missing tenant/security binding, unsupported trust roots, and attempted production-trust assertion.

The boundary output separates four concepts that could otherwise be conflated:

Concept	Meaning in cycle 29
Mechanical signature validity	The test envelope verifies as a fixture mechanism.
Policy admissibility	The policy document contains the required replacement fields and controls.
Trusted production attestation	Not granted by the fixture policy. Requires real deployment root evidence.
Production claim readiness	Not granted. Still requires intake, conformance, ingestion, security/provenance, noise floor, threshold replay, readiness, and handoff gates.

The cycle 29 auditor validated M-TRUSTPOL-1. The ledger records two scoped auditor fixes: collector firmware identity binding was enforced as part of collector identity binding, and replacement-map `input_present` cells were normalized as booleans. The auditor regenerated the artifacts and reported that the verifier, Python compilation, independent CSV/figure probe, `promise_check`, and `org_check` passed with only known root package warnings.

The cycle 29 auditor ledger event is `8fdf4980-60f4-4f13-b1f1-8d2a93693882`.

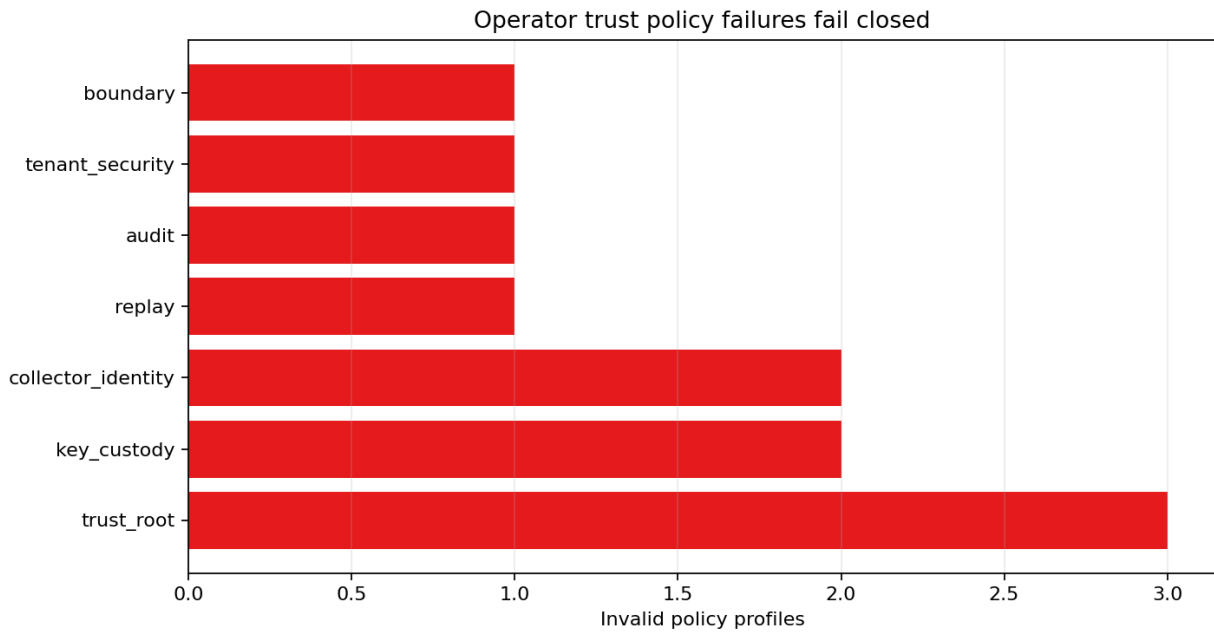


Figure 2: Fail-closed policy failures by trust root, key custody, collector identity, replay, audit, and tenant/security category.

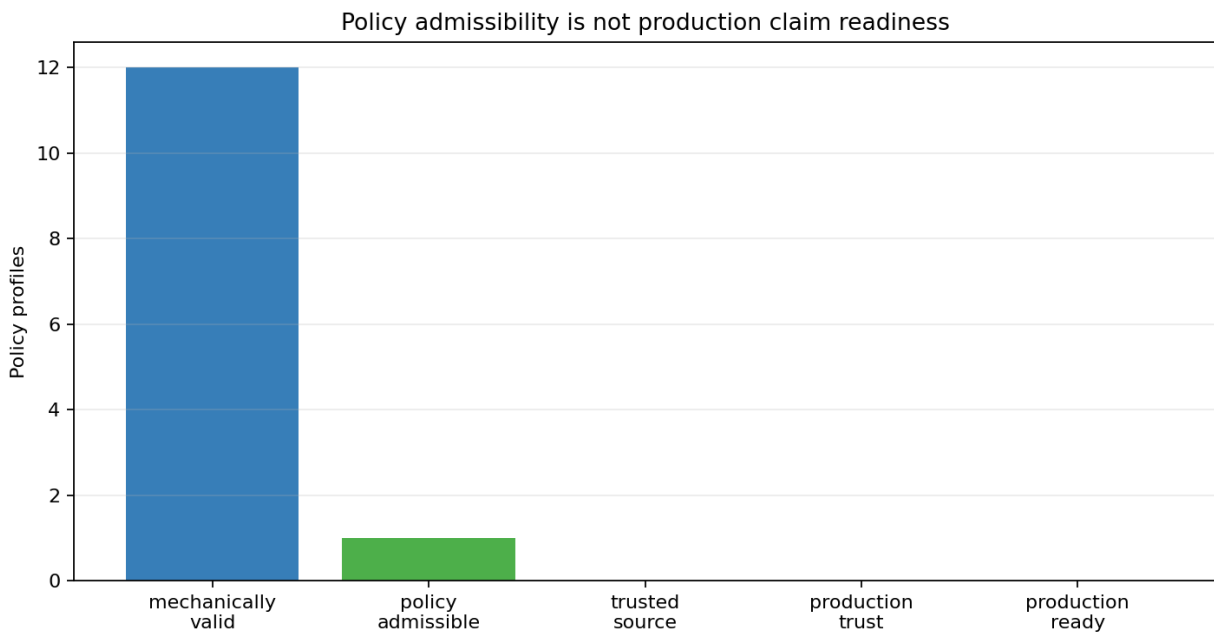


Figure 3: Boundary between mechanical signature validity, policy admissibility, trusted production attestation, and production claim readiness.

Cycle 30: End-to-End Evidence Gatechain Replay

Cycle 30 added M-GATECHAIN-1, an executable promotion-state replay. A gatechain is an ordered sequence of required states that evidence must pass before it can affect a production claim. The purpose is to prove that no earlier pass, no state presence alone, no fixture label, no skipped step, and no downstream shortcut can reach claim credit.

The gatechain contains 14 required states:

1. raw_bundle_seen
2. attestation_mechanically_valid
3. trust_policy_admissible
4. intake_structurally_admissible
5. adapter_conformant
6. adapter_normalized
7. production_ingestion_accepted
8. security_provenance_passed
9. noise_floor_passed
10. threshold_replay_passed
11. planner_update_eligible
12. final_readiness_update_eligible
13. handoff_traceable
14. production_claim_credit_allowed

The transition rules are linear. Adjacent states require identifier continuity for `bundle_id`, `measurement_run_id`, `operator_id`, `collector_id`, and `schema_version`. A skipped gate, out-of-order gate, unknown state, duplicated state, adjacent identifier mismatch, failed state, or invalid evidence label quarantines the path at the first affected transition with a named `blocked_reason`.

The validated gatechain outputs are:

Artifact	Rows	Role
data/evidence_gatechain_state_schema.csv	14	Required promotion states
data/evidence_gatechain_transition_rules.csv	14	Transition order, identifier-continuity rules, evidence-label rules, and missing-state reasons
data/evidence_gatechain_valid_fixture_paths.csv	26	Ordered fixture paths that exercise gate ordering but still quarantine before production claim credit
data/evidence_gatechain_invalid_fixture_paths.csv	202	Invalid path rows for skipped gates, out-of-order states, identifier mismatches, failed gates, evidence-label violations, and downstream bypass attempts
data/evidence_gatechain_replay_results.csv	19	Replay summaries

Artifact	Rows	Role
data/evidence_gatechain_quarantine_reasons.csv	17	Fail-closed quarantine reasons
data/evidence_gatechain_claim_credit_boundary.csv	19	Claim-credit boundary rows
data/evidence_gatechain_traceability_matrix.csv	5	Links to attestation, intake, adapter conformance, production ingestion, final readiness, and handoff artifacts

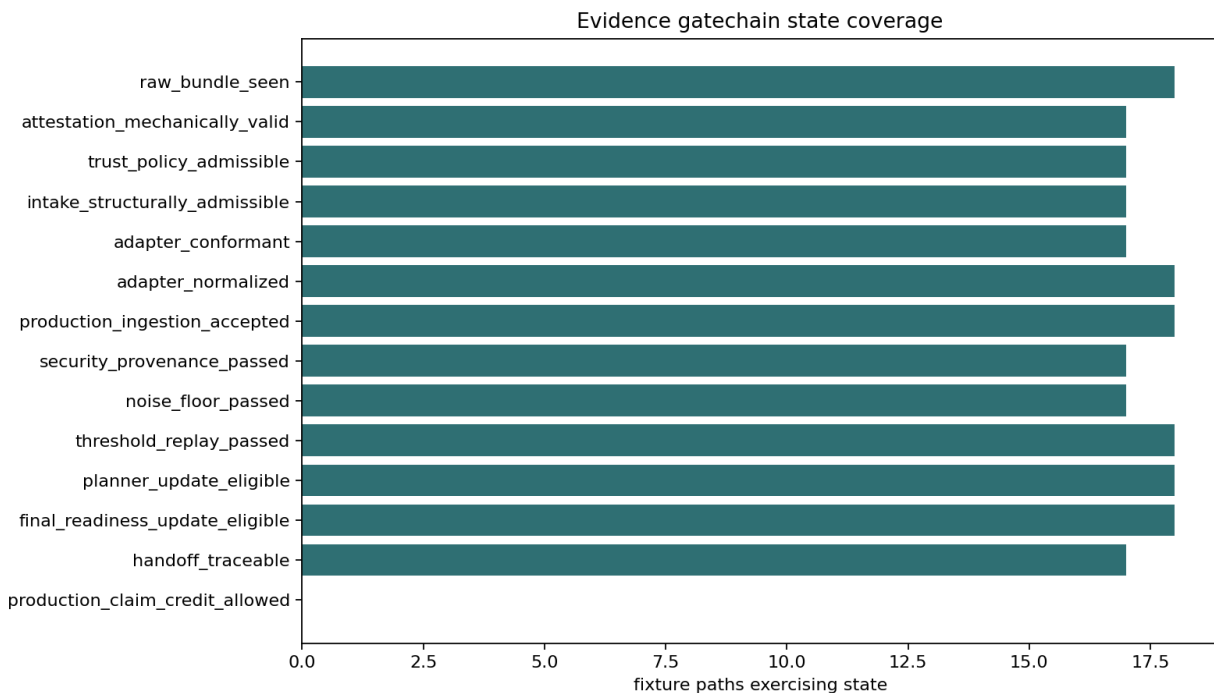


Figure 4: Required gatechain states and which fixture paths exercise them.

The replay covers both validly ordered fixture-boundary paths and invalid paths. Invalid paths include empty path, skipped attestation, skipped trust policy, skipped intake, skipped adapter conformance, out-of-order state, mismatched bundle ID, mismatched measurement run ID, mismatched operator ID, mismatched collector ID, mismatched schema version, fixture evidence at production ingestion, threshold replay without security/provenance, threshold replay without noise floor, proxy evidence threshold-credit attempt, final readiness without handoff, and failed attestation.

The supplied audit report found one critical defect in `scripts/replay_evidence_gatechain.py`. Replay had treated a gate as passed when its `state_id` appeared, ignoring `state_passed=false`. That meant a future `production_target` path with failed attestation could incorrectly reach `production_claim_credit_allowed=true`.

The applied fix changed replay semantics so a state must both appear and have `state_passed == true`. The worker added an `invalid-failed-attestation` fixture and verifier assertions for failed-

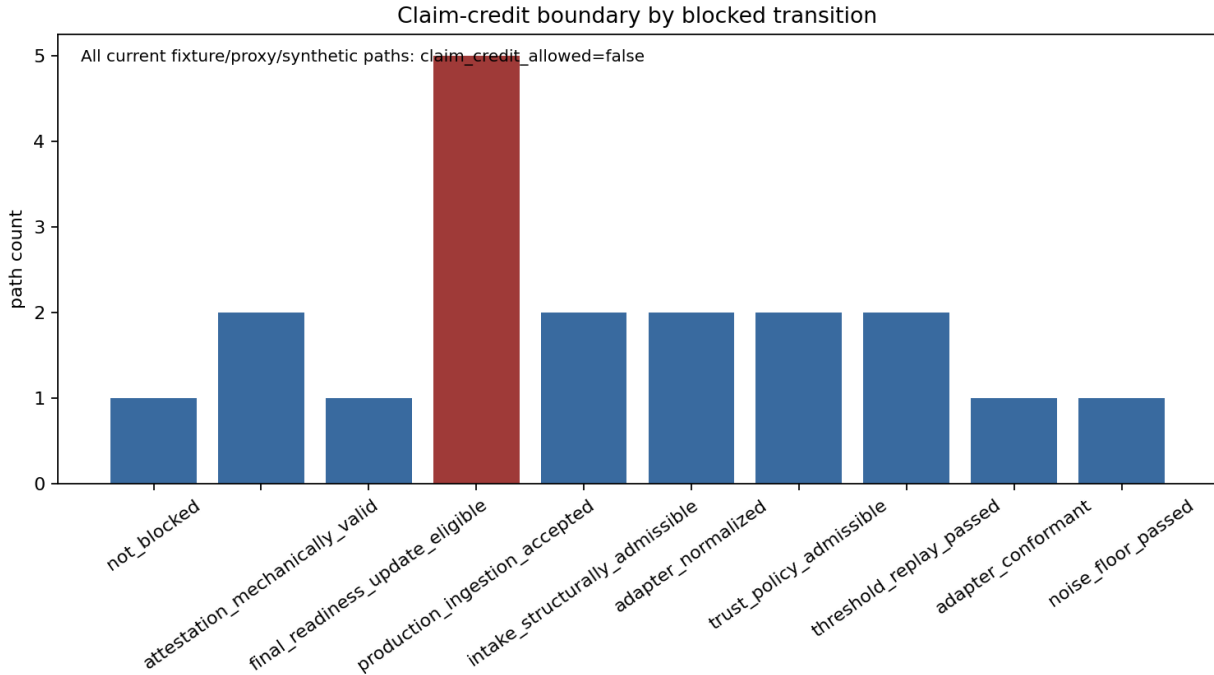


Figure 6: Allowed and blocked transitions from raw bundle through final readiness, highlighting that fixtures stop before production claim credit.

event. This report treats cycle 31 as a record gap and reporting boundary rather than a technical result.

This gap does not change the accepted state of cycles 29 and 30. It means only that there is no cycle-31 technical artifact to summarize from the available workspace record.

Discussion

Cycles 29-31 add two controls to the pre-production evidence path.

Cycle 29 answers a trust-root question: what must replace fixture signing before a telemetry bundle can be treated as production-origin evidence? The answer is a policy gate with non-exportable production key custody, access control, rotation, revocation, registered collector identity, software and firmware identity binding, topology binding, manifest and payload digest binding, schema and bundle registry binding, replay protection, tenant/security binding, retention authorization, append-only audit logs, verifier identity, and incident response ownership. A fixture policy can prove the required fields exist. It cannot prove production trust.

Cycle 30 answers a composition question: can individual gates be bypassed when their outputs are chained together? The accepted replay answer is no. The gatechain requires ordered state progression, identifier continuity, true state-pass values, production labels only at production-only states, security/provenance and noise-floor gates before threshold replay, and handoff traceability before final claim credit. The auditor's failed-state patch is important because it prevents the state machine from confusing "a gate produced a row" with "the gate passed."

Together, these cycles convert earlier one-gate validations into a more complete production promotion boundary. The package now has both individual gate validation and an executable

end-to-end replay. The boundary is still intentionally conservative: no fixture signature, trust-policy profile, adapter fixture, conformance pass, intake admission, synthetic trend, planned deployment, host-local proxy, failed gate, skipped gate, unjoined row, below-noise row, or security-denied row can grant production claim credit.

The highest-value next work remains real production telemetry or deployment-root integration evidence. The supplied audit guidance explicitly says to move on from M-GATECHAIN-1 and avoid adding another synthetic promotion layer unless it tests a new fail-closed boundary not already covered.

Conclusions and Recommendations

Cycles 29-31 produced two validated additions and one explicit record gap:

1. M-TRUSTPOL-1 defines the operator trust-policy gate required before fixture signing can be replaced by production KMS, HSM, hardware-attestation, or operator certificate-authority roots. It validates policy admissibility only; it does not grant production trust, production calibration, production readiness, or claim credit.
2. M-GATECHAIN-1 composes adapter, intake, attestation, trust-policy, ingestion, threshold, readiness, and handoff gates into one fail-closed promotion-state replay. After the auditor patch, failed states cannot pass by mere state presence. All 19 replay paths keep `production_claim_credit_allowed=false`.
3. Cycle 31 has no separate technical artifact in the available workspace record.

The recommended next technical step is actual trusted `production_target` telemetry or deployment-root integration evidence replayed through the full path:

- intake bundle admission
- operator trust-policy admission with real deployment roots
- attestation mechanics against real production trust material
- adapter conformance
- adapter normalization
- production ingestion
- security/provenance validation
- noise-floor checks
- DC-001/DC-002 threshold replay
- planner updates
- final readiness
- handoff traceability

The architecture stance remains unchanged. Option A remains the default for controls, zero-reuse, cheap-recompute, high-overhead, or unjoinable regimes. Option B and Option C remain contract-ready pathways only when production rows prove safe object reuse or trajectory/DAG value under joined bytes, power, latency, tenant, topology, security, provenance, retention, verifier, interval, and noise-floor gates.

References

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Appendix: Implementation Details

Code Organization

Cycle 29 added or updated the operator trust-policy package:

File	Purpose
<code>memory-centric-agentic/operator_trust_policy.md</code>	Operator note describing the fixture-to-production signing replacement boundary
<code>scripts/build_operator_trust_policy_fixtures.py</code>	Builds trust-policy schema, profiles, invalid cases, key lifecycle rows, and attestation replacement map
<code>scripts/evaluate_operator_trust_policy.py</code>	Evaluates policy admissibility, failure modes, production-trust boundary rows, and traceability
<code>scripts/plot_operator_trust_policy.py</code>	Renders policy coverage, failure-mode, and boundary figures

File	Purpose
tests/verify_operator_trust_policy.py	Verifies schema coverage, invalid profiles, production-trust boundaries, traceability, and figures

Cycle 30 added or updated the evidence gatechain package:

File	Purpose
memory-centric-agentic/evidence_gatechain.md	Operator note describing ordered promotion states and production claim-credit boundaries
scripts/build_evidence_gatechain_fixtures.py	Builds required states, transition rules, valid fixture paths, and invalid fixture paths
scripts/replay_evidence_gatechain.py	Replays promotion paths, enforces failed-state semantics, emits quarantine and claim-boundary outputs
scripts/plot_evidence_gatechain.py	Renders state coverage, quarantine-reason, and claim-boundary figures
tests/verify_evidence_gatechain.py	Verifies states, skipped/out-of-order failures, identifier mismatches, failed gates, boundaries, traceability, and figures

Figure Inventory

Cycle 29 figures:

Figure	Dimensions	Bytes
data/operator_trust_policy_coverage.png	1600 x 800	40,450
data/operator_trust_policy_failures.png	1440 x 768	41,358
data/operator_trust_policy_boundary.png	1440 x 768	40,281

Cycle 30 figures:

Figure	Dimensions	Bytes
data/evidence_gatechain_state_coverage.png	1600 x 928	99,084
data/evidence_gatechain_quarantine_reasons.png	1600 x 896	205,168
data/evidence_gatechain_claim_boundary.png	1520 x 864	119,297

Validation Results

Cycle 29 validation was recorded by auditor event 8fd4980-60f4-4f13-b1f1-8d2a93693882. The accepted validation regenerated:

- 23 trust-policy schema rows

- 1 complete profile
- 11 invalid profiles
- 5 key-lifecycle rows
- 10 replacement-map rows
- 12 policy results
- 8 failure categories
- 12 boundary rows
- 4 traceability links
- three nonblank figures

The auditor reported that tests/verify_operator_trust_policy.py, Python compilation, independent CSV/figure probe, promise_check, and org_check passed with only known root package warnings.

Cycle 30 validation was recorded by auditor event 7e4d5f28-f3c7-4ea5-8d0d-4467322f4f4a. The accepted validation regenerated:

- 14 state rows
- 14 transition rules
- 26 valid fixture rows
- 202 invalid fixture rows
- 19 replay summaries
- 17 quarantine reasons
- 19 claim-credit boundary rows
- 5 traceability rows
- three nonblank figures

The supplied audit report states that build_evidence_gatechain_fixtures.py, replay_evidence_gatechain.py, plot_evidence_gatechain.py, tests/verify_evidence_gatechain.py, Python compilation, an independent failed-gate reproducer, an independent CSV/figure probe, promise_check, and org_check passed. promise_check was green with 155 events and 29 plan milestones.

Source Session References

Cycle	Role	Session ID
29	researcher	2f10dd0e-410c-437f-86b7-632c6afbc58f
29	worker	cd141b85-6781-49f4-a0b1-79f46a87b759
29	auditor	f57a5389-0956-4b3e-b994-6fc4e5aeb85b
30	researcher	430d6071-c8ff-4d6d-919a-8955a2de14f5
30	worker	7b936d68-21d3-4a35-b928-04cab99f5417
30	auditor	d4f6088c-1a41-40ee-8003-49883038baa3
31	researcher	4e6ccae6-d9c1-468f-9c9f-f901bafabe2b
31	worker	58d60678-2d3b-41c2-a833-68738527836b
31	auditor	e53d1f06-83a9-4327-b2f1-5159718cae6b

Cross-Reference Map

Origin	Consuming artifact	Flow
data/production_attestation_results.csv	data/operator_attestation_replacement_map.csv, data/evidence_gatechain_state_schema.csv	Mechanical test attestation remains upstream evidence and must be replaced by real operator trust roots before production trust.
data/operator_trust_policy_results.csv	data/evidence_gatechain_valid_fixture_paths.csv, data/evidence_gatechain_transition_rules.csv	Policy admissibility becomes a required gatechain state but cannot by itself grant production trust.
data/production_intake_admission_results.csv	data/evidence_gatechain_valid_fixture_paths.csv, data/evidence_gatechain_traceability_matrix.csv	Intake admission is required before adapter conformance and production ingestion states.
data/adapter_conformance_results.csv	data/evidence_gatechain_valid_fixture_paths.csv	Adapter conformance is required before adapter normalization and production ingestion.
data/production_dcl2_ingestion_results.csv	data/evidence_gatechain_replay_results.csv, data/evidence_gatechain_claim_credit_boundary.csv	Production ingestion, security/provenance, noise-floor, and threshold gates are replayed before any claim-credit attempt.
data/final_claim_readiness_matrix.csv and data/handoff_claim_traceability.csv	data/evidence_gatechain_traceability_matrix.csv	Final readiness and handoff traceability are required downstream gates before claim credit.

Manifest Snapshot

MANIFEST.md was replaced with a current workspace snapshot after cycles 29-31. It reports:

Category	Count
Python scripts in scripts/	61
Wolfram scripts in scripts/	4
Test scripts	16
Total scripts/ lines	16,277
Markdown model/synthesis files under memory-centric-agentic/	29
Experiment-plan and verification markdown files	4
CSV data/model files under data/	157
Figures under data/	74
Sub-topics completed, assessed, or designed	29

No ## Key Files section was present in the prior manifest, so there was no final-reporter-owned section to preserve.